

Spark Practical Sessions

There are 3 practical session with Spark. The aim of these sessions is to practice with the Spark framework. You will use the interactive pyspark shell which is a Python 3 interactive terminal with a running Spark framework.

You will have to upload your work at the end of each practical session.

1 Todo before the practical session

If you are not familiar with python, practice the Python 3 tutorial before the practical session: <https://docs.python.org/3/tutorial/index.html>

If you want to install pyspark on your computer, do it **before** the practical session. And **test your installation** as explained in point 6 bellow.

2 Documentation

You will use Spark version 2.3 or higher. These versions provide another data structure (DataFrames) in addition to RDDs, but you will use RDD (RDD are less powerful but simpler).

The Quick Start guide for Spark version > 2.1 is mainly focused on DataFrames and not RDDs. Therefore, you will use the Quick start guide of version 2.0.

Open the Spark start page (version 2.0) in a browser: <https://archive.apache.org/dist/spark/docs/2.0.0/quick-start.html>. In this tutorial, each example is available in several languages (scala, python,...). You have to select the python version.

Also open in your browser the Spark documentation on RDDs: <http://spark.apache.org/docs/latest/rdd-programming-guide.html>. The sections about transformations and actions are those that you will need for the practical session.

3 Todo during the practical session

1. Read the section Documentation above.
2. If you have not installed Spark on your own computer, you have to log into a linux session on a university computer (you may have to reboot to select linux).
3. Download the **sessionX.zip** file from Moodle and unzip it (There are 3 practical sessions, so X is either 1,2 or 3).

This creates a directory **Practical_Session_X** in which you will find the **sessionX.txt** file and a **data/** directory containing several example data files.

4. Open a terminal in the directory **Practical_Session_1/** that was created when you unzipped session1.zip.
5. Now, you can start the pyspark shell in the terminal or a notebook:

- pyspark in a terminal: (This command works for those that use the university computers, if you have your own installation of Spark, you will have to adapt the path to pyspark binary).

```
/home/jeudbapt/local/spark-3.5.0-bin-hadoop3/bin/pyspark
```

This launches a pyspark shell. It may takes a few seconds to start.

- pyspark in a notebook: (This command works for those that use the university computers)

```
/home/jeudbapt/local/spark_notebook.sh
```

This should launch a new notebook tab in your browser.

If you have Spark on your computer, you can try:

```
PYSPARK_DRIVER_PYTHON="jupyter-notebook" spark_directory/spark-3.5.0-bin-hadoop3/bin/pyspark
```

where **spark_directory** is the path to the directory where you have installed Spark.

6. Then to try the pyspark shell, type (in the pyspark shell or the notebook):

```
sc.parallelize(range(50)).collect()
```

It should output the list [1,2,3,...,49]. Otherwise, pyspark is not working.

7. You can now open **sessionX.txt** in a text editor. The questions are in this file, and you will have to write your answers in it.
8. At the end of the session, upload this file **sessionX.txt** on Moodle (BIG DATA course), *even if you have not finished*. Do not upload a notebook, I need your answers in the file **sessionX.txt**.
9. If you have not finished at the end of the session, you can continue at home and upload another version up to a week later.